AI-Powered Agriculture Crop Advisory Assistant

Design Document — Architecture Decisions, Trade-offs & Rationale

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1. System Philosophy & Motivation

Agricultural decision-making is characterized by high uncertainty, regional heterogeneity, and interdependent ecological variables. Generic LLMs hallucinate numerical agronomic claims; isolated statistical models cannot interpret conversational queries. This system bridges the divide through a deterministic hybrid decision-support architecture.

2.x RAG (Retrieval-Augmented Generation)

Forces advisory responses to cite concrete historical cases. Prevents hallucinated yield numbers. Allows index updates without LLM retraining.

2.x FAISS for Vector Search

Sub-millisecond Cosine similarity on CPU. Self-contained portable binary index. No cloud dependencies.

2.x Hybrid Search (Dense + Sparse + RRF)

Dense captures semantic intent; Sparse captures exact agronomic tokens (pH 6.5, Kharif). RRF (k=60) fuses both rankings without score normalization artifacts.

2.x Metadata Filtering

Hard physical constraints (State, Season, Crop) enforced pre/post-retrieval to guarantee agro-ecological validity.

2.x Random Forest for Yield Prediction

No hallucinated numbers. Tree-ensemble variance gives exact 80% CI. <5ms inference. Handles non-linearities in tabular agronomy data.

2.x 8-Agent Deterministic Pipeline

Separation of concerns. Sub-100ms E2E. Reproducible. Each agent has a strict typed contract.

2.x Multi-tier Guardrails

Input injection defense + domain scope + agronomic bounds. Output disclaimer injection. LLM Judge groundedness verification.

2.x FastAPI + Streamlit

FastAPI: async ASGI, Pydantic v2, auto OpenAPI docs. Streamlit: rapid interactive frontend with full API integration.

3. Engineering Trade-offs Matrix

Dimension	Selected	Alternative	Rationale
Vector DB	FAISS (Local)	Pinecone / Qdrant	Zero cloud cost, <2ms latency, portable binary
Search	Hybrid Dense+Sparse	Pure Vector	Exact agronomic token matching preserved
Yield Model	Random Forest	Neural Net / LLM	Bounded, non-hallucinatory, tree-variance CI
Orchestration	Deterministic Pipeline	ReAct Agent Loop	<100ms guaranteed, reproducible execution
Deployment	Local-first	Cloud Serverless	Rural offline resilience, zero infra cost
Frontend	Streamlit + HTML SPA	React / Next.js	Rapid iteration, full Python ecosystem access

4. Failure Modes & Graceful Degradation

- Missing OpenAI Key → local template synthesis; all search/ML features remain functional
- Missing raw dataset → auto-generates 1,500 synthetic agronomic cases
- Out-of-distribution inputs → Risk Agent flags 3-sigma deviations
- Empty retrieval → hierarchical metadata relaxation (State→Season→Crop)